

What this is

A controller that drives the carriage and the cross slide with stepper motors, keeping them in step with the spindle. It does the job of change gears and a lead screw, and adds automatic operations on top: mark the corners of the cut and it will make the passes for you.

|                 |                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Limits          | Most jobs need you to mark the corners of the cut first — how far along, and how deep in — with the four limit keys. They are not guard rails: they are the cut. |
| Passes          | A cut taken in stages. You say how many; it divides the depth between them and lifts the tool clear on the way back.                                             |
| Feed            | How far the tool moves per turn of the spindle. On a thread that is the pitch; on a turning cut it sets the finish.                                              |
| It moves itself | Keep clear, know where the stop key is, and try anything new with the tool well away from the work.                                                              |

Choosing what it does

Each job has a key, shown in the table below. Two keys have a second job behind them: press the same key again to reach it, marked ×2.

**M** steps through the rest: **fourth axis** (if fitted) → **ball and dish** → **G-code** → **steady feed** → **slotting** → back to the gearbox

| Job            | Shows | Key | What it is for                                                                          | Limits     | What you can adjust                                             |
|----------------|-------|-----|-----------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------|
| Gearbox        | —     |     | The carriage feeds along by itself as the spindle turns — what replaces change gears.   | not needed | —                                                               |
| Cross-feed     | XGEAR | ×2  | The same, driving the cross slide across the face. Powered facing.                      | not needed | —                                                               |
| Turning        | TURN  |     | Reduces a diameter over several passes, lifting the tool clear on each return.          | all four   | Passes · Spring passes · Clearance                              |
| Facing         | FACE  |     | Cleans up or shortens the end of the work, in several passes.                           | all four   | Passes · Spring passes · Clearance                              |
| Parting        | CUT   |     | Cuts the work off, or cuts a groove. Can back out to clear the swarf.                   | up, down   | Passes · Peck depth · Clearance                                 |
| Threading      | THRD  |     | Cuts a screw thread, deeper each pass. Pick it from the list of 52 and the feed is set. | all four   | Passes · Spring passes · Flank infeed · Clearance · Thread list |
| Tapered thread | TPR   | ×2  | A thread on a taper rather than a parallel bar — pipe fittings, NPT and BSPT.           | all four   | as threading, plus its own taper                                |
| Taper          | CONE  |     | The tool moves in as it travels along, at a ratio you set. Feed in by hand.             | not needed | taper ratio, asked at the start                                 |
| Ball and dish  | ELLI  |     | Rounds the end into a ball, or hollows it into a dish.                                  | all four   | Passes                                                          |
| Slotting       | SLOT  |     | The lathe as a shaper: spindle still, tool stroking. Keyways and flats.                 | all four   | Passes · Stroke shortening                                      |
| Steady feed    | ASY   |     | A set feed rate with the spindle out of it. Handy for polishing.                        | not needed | —                                                               |
| G-code         | GCODE |     | Runs a program written on a computer and sent over.                                     | not needed | —                                                               |
| Fourth axis    | A1    |     | Drives a fourth motor, if you have fitted one.                                          | not needed | —                                                               |

Running one of the automatic jobs

|   |  |                                                                                                                                                                                                                    |
|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 |  | Set the feed, then move the tool to each corner of the cut and press the matching limit key. On a thread the <b>up</b> limit is full thread depth — see below.                                                     |
| 2 |  | The bottom line reads <b>ON to set up 3 steps</b> . Press play to begin.                                                                                                                                           |
| 3 |  | It asks a short series of questions. <b>1/3</b> tells you which one you are on and how many there are. <b>&lt;&gt;</b> means the arrow keys change the answer; otherwise type a number. Play accepts and moves on. |
| 4 |  | The last question is <b>Go?</b> , and it shows how far the tool will travel to reach the starting corner before it cuts anything. Play starts the job; stop takes you back to the beginning.                       |
| 5 |  | While it runs the line reads <b>Pass 2 of 6</b> . Play skips to the next pass if you have taken enough off; stop, or any jog key, ends the job.                                                                    |

**How deep should a thread go?** The controller sets the pitch from its thread list, but it cannot know the depth — that is what the up limit tells it, and getting it wrong is the usual reason a first thread comes out badly. For an ordinary 60° thread the depth is **0.6134 × pitch** measured inwards from the surface, so an M10×1.5 goes 0.92mm deep. If the cross slide is set to read diameters rather than distance from centre, put in twice that.

Reading the screen

|                     |
|---------------------|
| THRD* off step 1.00 |
| Pitch 1.50mm x2     |
| Z 12.345 X 25.400   |
| 3/3 Go Z-5.00mm?    |

|        |                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| Top    | Which job is selected, whether it is running, a small mark for each limit you have set, and how far one tap of a jog key moves.       |
| Second | The feed, and the number of starts if the thread has more than one.                                                                   |
| Third  | Where the tool is. A slashed X means that figure is a diameter.                                                                       |
| Bottom | Everything else — the setup questions, how many passes are done, why something was refused, and whichever readout you have turned on. |

The **display** key changes what the bottom line shows when it is otherwise idle: spindle angle, then spindle speed alongside the cutting speed at the tool, then big position digits you can read from a few feet away, then nothing.

## Setting up a new machine

Flash the firmware, then choose millimetres or inches with the measure key. The controller has to be told about your lathe before it can be accurate, and it can measure most of that itself. **Hold** the settings key, choose **Calibration**, press play, and work down this list — it begins with the routines that move nothing and ends with the one that deliberately stalls a motor.

| # | Routine            | Does it move?      | What it is for                                                                                     |
|---|--------------------|--------------------|----------------------------------------------------------------------------------------------------|
| 1 | Input tester       | nothing moves      | Press every key and check it registers. Confirms the panel works before anything is asked to move. |
| 2 | Encoder signal     | nothing moves      | Watches the spindle sensor for electrical interference. Should sit at 100 while the spindle turns. |
| 3 | Encoder PPR        | you turn the chuck | Turn the spindle a set number of times by hand; it works out how finely the sensor reads.          |
| 4 | Encoder direction  | nothing moves      | Turn the spindle forwards. If the number counts down instead of up, this flips it.                 |
| 5 | Z / X direction    | the axis moves     | Checks each axis travels the way you expect, before anything is measured against it.               |
| 6 | Z / X screw pitch  | the axis moves     | Moves a set distance; you measure what it actually moved, and it corrects itself to match.         |
| 7 | Z / X backlash     | the axis moves     | Finds the slack in each screw against a dial indicator, so it can be taken up automatically.       |
| 8 | Z / X travel limit | you jog it         | Wind each axis to its ends so the controller knows how far it is allowed to go.                    |
| 9 | Z / X max speed    | the axis moves     | Speeds up until the motor stalls, then backs off. Leave this one until last.                       |

Then save what you have done — there is no undo. Connect a computer over USB, send a single \$, and it prints every setting as a list of lines you can keep in a text file. Paste them back to restore. The same file can be downloaded from the web page.

## Settings that describe your lathe

**Hold** the settings key to open the menu. Everything here describes the machine itself, and you will mostly set it once. Anything to do with one particular job — how many passes, how much clearance — lives on that job's own page instead, which is a **short** press of the same key.

| How things behave       |                                                                                                                                                  | Each axis · Z, X and the fourth |                                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------|
| X readout               | Whether the cross slide shows how far it is from centre, or the diameter that gives — twice as much. Lathe work is usually quoted as a diameter. | Invert direction                | If the axis travels the wrong way.                                                |
| Retract distance        | How far the tool jumps clear when you press the retract key.                                                                                     | Backlash                        | The slack in the screw, taken up automatically whenever it reverses.              |
| Manual step time        | How long one tap of a jog key takes.                                                                                                             | Lead screw pitch                | How far the axis moves for one turn of its screw.                                 |
| Step rest               | A pause between stepped moves.                                                                                                                   | Motor steps/rev                 | From the motor and its driver settings together.                                  |
| Spindle divisions       | Turns the angle display into an indexer for cutting flats or keyways. 0 is off; 6 gives you a hexagon.                                           | Motor / screw pulley            | Tooth counts, if the motor drives through a belt.                                 |
| Index tolerance         | How close to a mark counts as being on it.                                                                                                       | Start speed                     | How gently it sets off before speeding up.                                        |
| Constant speed          | The single switch for the cutting-speed helper below.                                                                                            | Max speed                       | The fastest it will travel.                                                       |
| Material                | What you are cutting. Sets a sensible cutting speed for you.                                                                                     | Acceleration                    | How briskly it gets up to that speed.                                             |
| Tool                    | High speed steel or carbide. Carbide takes roughly three times the speed.                                                                        | Max travel                      | How far the axis can physically go, used as a safety net.                         |
| Manual speed            | Your own figure, used when Material is set to Manual.                                                                                            | Hold when idle                  | Whether the motor stays locked when stopped. Turn off if it runs hot.             |
| Spindle max rpm         | The fastest your lathe runs, so it never suggests more.                                                                                          | Fitted / Rotary                 | Fourth axis only: whether it exists, and whether it turns or slides.              |
| Spindle sensor          |                                                                                                                                                  | Handwheels                      |                                                                                   |
| Encoder PPR             | How many pulses it gives per turn. Usually printed on the body.                                                                                  | Fitted 1 / 2                    | Whether each handwheel is connected.                                              |
| Direction               | Flips the counting direction without rewiring anything.                                                                                          | Drives 1 / 2                    | Which axis it winds.                                                              |
| Spindle / sensor pulley | Tooth counts, if the sensor is belt driven rather than straight off the spindle.                                                                 | Invert 1 / 2                    | If it winds the wrong way.                                                        |
| Divider                 | Ignores very small movements, for a sensor that twitches at rest.                                                                                | Pulses per rev                  | Of the handwheel itself.                                                          |
| Dead-band               | How far the spindle must turn back before the tool follows it.                                                                                   | Min pulse width                 | Ignores electrical noise on its wiring.                                           |
| Dead-band shape         | One-way suits a worn lead screw; symmetric suits a noisy sensor.                                                                                 | Dead-band                       | Stops a wobble being read as a change of direction.                               |
| Glitch filter           | Ignores electrical spikes. Set too high it starts missing real pulses.                                                                           | Joystick                        |                                                                                   |
|                         |                                                                                                                                                  | Fitted                          | Uses all six spare terminals, so nothing else can share them.                     |
|                         |                                                                                                                                                  | Debounce                        | Ignores the brief chatter a switch makes as it closes.                            |
|                         |                                                                                                                                                  | WiFi                            |                                                                                   |
|                         |                                                                                                                                                  | Enabled                         | Makes the controller its own small network. Off means the radio is not on at all. |
|                         |                                                                                                                                                  | Access PIN                      | The password, exactly 8 digits. Change it from the default.                       |

Every thread it knows

Choose one from the machine and the feed is set for you. **Depth** is how far in from the surface the tool has to go, in millimetres, measured on the radius — that is what the up limit is for, and it is the one thing the controller cannot work out for itself. If the cross slide is set to read diameters, put in twice the figure shown.

| Thread           | Pitch  | Depth | Thread                      | Pitch | Depth | Thread                          | Pitch  | Depth |
|------------------|--------|-------|-----------------------------|-------|-------|---------------------------------|--------|-------|
| Metric · 60°     |        |       | Unified · UNC and UNF · 60° |       |       | British parallel pipe · 55°     |        |       |
| M3 coarse x0.5   | 0.50mm | 0.31  | 1/4-20 UNC                  | 20tpi | 0.78  | G1/8 BSPP 28tpi                 | 28tpi  | 0.58  |
| M4 coarse x0.7   | 0.70mm | 0.43  | 1/4-28 UNF                  | 28tpi | 0.56  | G1/4 BSPP 19tpi                 | 19tpi  | 0.86  |
| M5 coarse x0.8   | 0.80mm | 0.49  | 5/16-18 UNC                 | 18tpi | 0.87  | G3/8 BSPP 19tpi                 | 19tpi  | 0.86  |
| M6 coarse x1.0   | 1.00mm | 0.61  | 5/16-24 UNF                 | 24tpi | 0.65  | G1/2 BSPP 14tpi                 | 14tpi  | 1.16  |
| M8 coarse x1.25  | 1.25mm | 0.77  | 3/8-16 UNC                  | 16tpi | 0.97  | G3/4 BSPP 14tpi                 | 14tpi  | 1.16  |
| M8 fine x1.0     | 1.00mm | 0.61  | 3/8-24 UNF                  | 24tpi | 0.65  | G1 BSPP 11tpi                   | 11tpi  | 1.48  |
| M10 coarse x1.5  | 1.50mm | 0.92  | 7/16-14 UNC                 | 14tpi | 1.11  | Trapezoidal · 30°               |        |       |
| M10 fine x1.25   | 1.25mm | 0.77  | 1/2-13 UNC                  | 13tpi | 1.20  | Tr8x1.5 trap                    | 1.50mm | 1.00  |
| M12 coarse x1.75 | 1.75mm | 1.07  | 1/2-20 UNF                  | 20tpi | 0.78  | Tr10x2 trap                     | 2.00mm | 1.25  |
| M12 fine x1.5    | 1.50mm | 0.92  | 5/8-11 UNC                  | 11tpi | 1.42  | Tr12x3 trap                     | 3.00mm | 1.75  |
| M14 coarse x2.0  | 2.00mm | 1.23  | 5/8-18 UNF                  | 18tpi | 0.87  | Tr16x4 trap                     | 4.00mm | 2.25  |
| M16 coarse x2.0  | 2.00mm | 1.23  | 3/4-10 UNC                  | 10tpi | 1.56  | Tr20x4 trap                     | 4.00mm | 2.25  |
| M16 fine x1.5    | 1.50mm | 0.92  | 3/4-16 UNF                  | 16tpi | 0.97  | ACME · 29°                      |        |       |
| M20 coarse x2.5  | 2.50mm | 1.53  | 1-8 UNC                     | 8tpi  | 1.95  | 1/4-16 ACME                     | 16tpi  | 1.05  |
| M24 coarse x3.0  | 3.00mm | 1.84  |                             |       |       | 5/16-14 ACME                    | 14tpi  | 1.16  |
|                  |        |       |                             |       |       | 3/8-12 ACME                     | 12tpi  | 1.31  |
|                  |        |       |                             |       |       | 1/2-10 ACME                     | 10tpi  | 1.52  |
|                  |        |       |                             |       |       | 5/8-8 ACME                      | 8tpi   | 1.84  |
|                  |        |       |                             |       |       | 3/4-6 ACME                      | 6tpi   | 2.37  |
|                  |        |       |                             |       |       | 1-5 ACME                        | 5tpi   | 2.79  |
|                  |        |       |                             |       |       | American taper pipe · 60° taper |        |       |
|                  |        |       |                             |       |       | 1/8-27 NPT                      | 27tpi  | 0.75  |
|                  |        |       |                             |       |       | 1/4-18 NPT                      | 18tpi  | 1.13  |
|                  |        |       |                             |       |       | 3/8-18 NPT                      | 18tpi  | 1.13  |
|                  |        |       |                             |       |       | 1/2-14 NPT                      | 14tpi  | 1.45  |
|                  |        |       |                             |       |       | 3/4-14 NPT                      | 14tpi  | 1.45  |

Depths are for a full-form thread on the outside of a bar. An internal thread is cut to the same depth outwards from the bore. Trapezoidal and ACME include the usual root clearance; taper pipe threads are deeper than they look because the form is truncated rather than pointed.

Cutting speeds

Metres per minute at the surface of the work — what the controller uses when you switch the cutting-speed helper on. Conservative starting points for turning, not limits: depth of cut, how rigid the setup is and whether you are using coolant all matter too. Work up from these if the finish and the swarf look happy.

| Material   | HSS      | Carbide   | Material    | HSS       | Carbide   |
|------------|----------|-----------|-------------|-----------|-----------|
| Aluminium  | 70 m/min | 200 m/min | Alloy steel | 20 m/min  | 90 m/min  |
| Brass      | 60 m/min | 180 m/min | Tool steel  | 15 m/min  | 60 m/min  |
| Bronze     | 40 m/min | 120 m/min | Stainless   | 15 m/min  | 60 m/min  |
| Copper     | 50 m/min | 150 m/min | Titanium    | 10 m/min  | 40 m/min  |
| Cast iron  | 25 m/min | 90 m/min  | Plastic     | 100 m/min | 250 m/min |
| Mild steel | 30 m/min | 120 m/min |             |           |           |

**Turning it into a spindle speed:** rpm = 318 × speed ÷ diameter in mm. So mild steel with carbide at 120 m/min on a 40mm bar wants about 950 rpm. The controller does this for you and keeps doing it as the diameter changes, which is what makes it useful when facing towards the centre.

Using it from a phone

|   |                                                                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Settings → WiFi → <b>Enabled</b> = yes. Change the <b>Access PIN</b> from the default first — anyone who can join the network can change the machine's settings.                                    |
| 2 | Join the <b>NanoEls-H4</b> network with that PIN, then open <b>192.168.4.1</b> . The panel shows the address.                                                                                       |
| 3 | Every setting is there, quicker to type than on the panel, with position and speed live. It also shows how clean the spindle sensor's signal is — leave it open through a job and check afterwards. |
| 4 | New firmware installs from the same page, with no computer at the machine. Only while it is stopped.                                                                                                |